

Do I Prefer It?: The Role of Cultural Heuristics in Chinese Citizens' Attitudes to COVID-19 Rumors

Han Shao

Renmin University of China

Beijing, P. R. China

shaohan0430@ruc.edu.cn

ABSTRACT

From the perspective of the prevalence of social media rumors, we put forward the concept of cultural heuristic and explore how it influences the attitude and behavior of the Chinese citizens towards rumors when facing epidemic again, i.e., COVID-19. We recruit 12 Chinese citizens to conduct semi-structured interviews, and use grounded theory to analyze the data coding. The results show that China's unique cultural, social and historical background and collective behavior convey the attitudes and beliefs to rumors thus lead to behavior. This study is an exploration of rumor culture in non-Western society. It can help us have a better understanding of the cultural cognition and value orientation between Chinese citizens and rumors.

CCS CONCEPTS

• **Human-centered computing** → *HCI theory, concepts and models; Computer supported cooperative work; Social media.*

KEYWORDS

social computing, online public opinion, privacy and security, collective intelligence, crisis informatics

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1 INTRODUCTION

Since the outbreak of novel coronavirus (COVID-19) in 2020, misinformation or rumors have become increasingly prevalent with the popularity of online social networks (OSN). The World Health Organization (WHO) considers COVID-19 as a "mass information disease" [23].

In order to explore whether and how the cultural factors affect the perception and behavior of social media rumors when Chinese citizens facing public health emergencies again, we propose the concept of cultural heuristics. Different from previous studies, we

focus on Chinese citizens. The reasons are as follows: (1) COVID-19 was first widely reported and paid attention to in China. It means that Chinese people do not have enough knowledge of novel coronavirus in this particular period. It is so typical that related research can most truly reflect citizens' attitude towards social media rumors. (2) China has experienced a similar epidemic outbreak of SARS virus in 2003, and citizens have certain memory and historical experience as the culture cognitive basis of rumors. (3) However, social media in 2003 was far less widely used than today, thus the focus of this article is about the rumors posted on contemporary popular social media. (4) Another unique point is that in China's epidemic prevention and control process, medical experts have adopted a combination of traditional Chinese and western medicine treatment. As a traditional culture with a long history and a unique cultural background in China, it may become one of the factors affecting citizens' attitudes. Although there have been some early epidemiological and clinical studies, few have examined the Chinese public's awareness and response to the epidemic. Therefore, it is valuable and interesting to explore whether and how the unique historical and cultural background of China influence citizens' judgment towards rumor in the face of large-scale epidemic again. This study is an exploration of rumor culture in non-Western societies and a supplement to existing work.

2 RELATED WORK

2.1 Psychological and Technical Rumor Research

Rumor research was first conducted in the field of social psychology. The classic definition of rumors is about ambiguity. There are four key factors affecting the spread of rumors: anxiety, uncertainty, relevant importance and belief [26]. Rumors are a common feature of crisis events [28]. The crisis environment is characterized by extreme uncertainty and lack of information, which all contribute to the emergence of rumors. Computer technology is also widely used to identify fake news and rumors. Different fields of research have used artificial intelligence algorithms, cognitive psychology and mathematical models to identify false information. Related researches focused on theories of mathematics, natural language processing, artificial intelligence, machine learning, deep learning, data science and human psychology to solve the problem of authenticity and credibility analysis [32] [35].

2.2 CHI and CSCW Rumor Research

Although the technology field has made great progress in understanding and controlling the spread of rumors, the rumor problem

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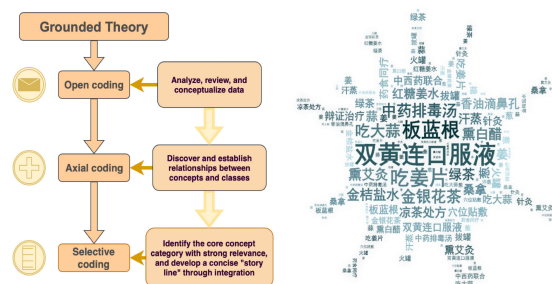


Figure 1: Three stages of grounded theory (left) and the top words related to traditional Chinese medicine in COVID-19 epidemic news on social media in January 2020 (right), including Shuanghuanglian, honeysuckle tea, herbal tea, moxibustion, acupuncture, eating garlic, fumigating vinegar, etc.

remains serious and difficult to control. CHI and CSCW communities have extensively discussed the use of rumors as part of online social public opinion from a human factor perspective. Researches include analysis of false information on the social media question [15], how Chinese Internet users use Weibo to cope with major disasters [25], Chinese Internet users' trust in the social media and official media news [34], how social media users correct crisis event online rumors [4], the role of formal channels in shaping the rumor spread [3], the origin and spread of information on Twitter [19], the importance of transmission dynamics [5], the role of journalists in the dissemination of misinformation [27] and so on. From the perspective of difference, numerous studies include the spread of rumors and the reactions of different groups to rumors [16], different reactions of Twitter users caused by different types of crises [22], and several quantitative measures are identified to constitute multidimensional characteristics [18].

2.3 Grounded Theory

Grounded theory is increasingly welcomed and used by CHI and CSCW community researchers [20]. As a scientific and effective method in qualitative research, it has been proved to be an effective research method in the fields of sociology, education and management [1]. The detailed process is shown in the Figure 1. CHI is facing more qualitative research issues to integrate with human values. Grounded theory has a wide range of applicability to CHI. Previous grounded theory research in CHI topics include online security [21] and password use [14]. It is also more suitable for contemporary human-computer interaction research. It goes beyond the coding process and allows people to create understanding with users. Related subjects include the crisis's impact on information sharing behavior [13], using the existing human-computer interaction theory to explore and elaborate the research results [10] and so on.

2.4 Heuristics from Different Perspectives

Heuristics is also widely discussed in the CHI community. Users are "cognitive misers" and often make disclosure decisions based on cognitive inspiration or psychological shortcuts (i.e., rules of

3 METHODS

In this section, we discuss the process of participant recruitment, summarize the data generation process, and briefly introduce how to use grounded theory to guide methods and build models, so that we can organize interview records, discover patterns and make targeted analysis.

3.1 Participant Recruitment

Due to the strict protection and isolation of most communities during the COVID-19 epidemic period, our field experiment is advertised through online social media, which has been widely used in related research. The platforms include WeChat and Weibo, which are the most widely used social media in China. The experimental date is from February 1 to February 7, 2020. In order to ensure the diversity of the sample, 12 active social media users aged between 20 and 55 were recruited. Each interviewee is identified by assigning a unique ID (P1-P12). Since the name of Chinese citizens consists of the first name and the last name, we will only use the ID and the last name to uniquely identify and protect the anonymity. As the content of the interview may involve personal privacy, participants are told that they can quit at any time and get more information from the researchers.

3.2 Data Generation

Data collection included an initial demographic questionnaire and a recorded semi-structured interview. After obtaining informed consent, we conducted semi-structured interviews. Most of the participants were interviewed through social software, telephone or instant messaging. The interview lasted 20-30 minutes, and all the content was recorded. In addition, some participants chose to review their social media history or shopping records to evoke details of their personal experiences as memory aids.

The experiment was set as a semi-structured interview. We planned several questions in advance, including some questions related to China's unique history and culture as heuristics. Then we conducted the interview accordingly, encouraging participants to speak freely and carry out further communication. The main questions include:

- (1) How does the COVID-19 epidemic affect your life?



Figure 2: People lined up to buy traditional Chinese medicine such as Shuanghuanglian(left) and pharmacies posted signs that they were out of stock(right). The images are from Weibo, one of the most popular social media platforms in China.

(2) Do you have any impression on the outbreak of SARS virus in 2003? How do you think the background at that time was different from COVID-19?

(3) How do you feel about the large number of news reports published on social media every day?

(4) Have you bought medicine such as Shuanghuanglian and Isatidis root (both are Chinese patent medicine) that have been widely reported on social media? Top words about traditional Chinese medicine in social media reports are shown in the figure 1.

3.3 Analysis and Interpretation of Data

The core elements of grounded theory provide an appropriate research process for studying the effects of catastrophic epidemic events on people's social media attitudes and behaviors. It is necessary and feasible to apply the grounded theory method to study the evolution process and influencing factors of social risks induced by network rumors [12]. According to the obtained interview data, we follow the grounded theory proposed by Charmaz [7]. We use grounded theory process to analyze transcriptional interviews and qualitative surveys, following three stages of open coding, axial coding and selective coding. We complete initial text analysis and keyword extraction through NVivo in order to eliminate subjective factors and unscientific classification. The following coding process was completed by three researchers collectively so as to ensure the theoretical sensitivity and reduce personal bias.

4 FINDINGS AND RESULTS

Grounded theory allows us to organize interviews and find patterns from the data. Consequently, we can gain a comprehensive, potentially valid, previously unknown, and ultimately understandable knowledge. In the following, we will present some analysis examples of the valuable content of the semi-structured interview results, and discuss the influence of the cultural heuristics on the attitudes and behaviors of Chinese citizens.

4.1 Social Media for Real-time News and Work

With the rapid development of OSN, social media has become a new way of rumor spreading through the Internet. Communication media has reconstructed people's way of life. Extensive user base lays the foundation for information acquisition and dissemination. Interviewees of different ages and life stages have different views and experiences. Wang(P2), a graduate student, describes the impact of the COVID-19 on his study and life:

"During the epidemic, I stay at home almost all the time because I can't go out and meet someone. The school's curriculum has been switched to online teaching. Although the Internet has made learning convenient, I still prefer to study in classroom because it makes me more focused."

At the same time, many companies began to work from home. Sun(P5) is a company employee who worked from home and had meetings through the Internet during the epidemic. He mentions:

"Working at home as a result of the outbreak has allowed me to spend more time on social media. The Internet is amazing. We can work and communicate anytime and anywhere."

For Hou(P6), who is relatively older, she has a deep memory of the SARS virus epidemic and talks about the difference between past and today:

"When the SARS epidemic broke out, I could only get relevant information from TV and newspapers. That means it was difficult for me to get timely information about the epidemic. Now the times has changed. I can always get the latest news and reports through social media, such as Weibo and WeChat, which release a large number of epidemic news reports and experts' speeches every day..."

4.2 Chinese Medicine as a Cultural Heuristic

For a specific rumor that has been refuted so far, we further investigate how cultural heuristics affect respondents' reactions and opinions. The rumor about Shuanghuanglian oral liquid which is a kind of Chinese patent medicine drug, was published on the official Weibo accounts of the People's Daily, the Global Times and other media on January 31. The iconic title reads "New Discovery: Shuanghuanglian Oral Liquid Can Inhibit Novel Coronavirus". However, the next day, official media of CCTV news denied the rumor, explaining that traditional Chinese medicines such as honeysuckle and forsythia had broad-spectrum antiviral effects and that there was no formal clinical study of COVID-19.

Because through news reports, people know that in the treatment process, experts mostly adopt a combination of Chinese and western medicine treatment. The analysis shows that most of the interviewees (P1,P2,P5,P8,P10,P11,P12) talked about their attitudes towards Chinese medicine drugs and treatments:

"Traditional Chinese medicine is the essence of China's medical science and intangible culture. The main ingredients of Shuanghuanglian are herbs and ingredients of traditional Chinese medicine, so I would prefer that the news reported by the media is a reliable reference."

"I also participated in the process of panic buying of Shuanghuanglian. I went to a lot of offline pharmacies trying to buy it. People's rush buying was too crazy. They could line up such a long line at the door of the drugstore in the cold night, which reminds me of the phenomenon that everyone rushed to buy Isatidis root crazily when the SARS virus broke out years ago."

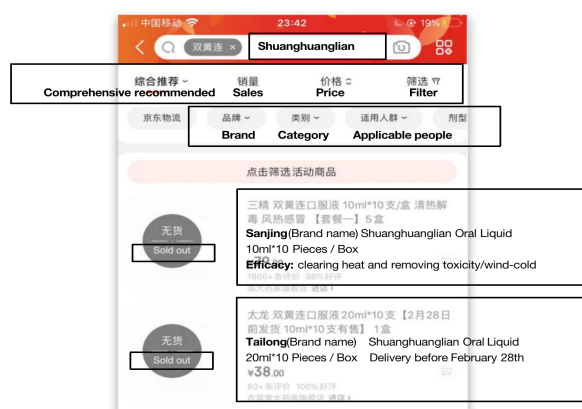


Figure 3: Shuanghuanglian and other traditional Chinese medicine also sold out rapidly on the JD.COM online shopping malls, one of the most popular e-commerce platforms in China.

Liu(P10) elaborates from the perspective of the connection and difference between Chinese and western medicine:

"Traditional Chinese medicine is about restoring the body's own ecological balance by stimulating the body's systems to fight off viruses and cure diseases. Western medicine, on the other hand, deals with virus, bacteria, and damages body's tissues and organs. The two are fundamentally different. I think they can complement each other. Traditional Chinese medicine focuses on individual differences and applies the right medicine to the patient's situation, which can largely make up for the shortcomings of western medicine."

Zhao(P11) gives an example from the perspective of his own experience:

"Personally, I have a strong belief in traditional Chinese medicine. Take my own experience as an example, I had western medicine treatment for allergic rhinitis for many years, but the effect was temporary. Later, I used traditional Chinese medicine with a try attitude. Now my rhinitis seldom relapses. So I believe traditional Chinese medicine has anti-virus effect."

4.3 Broader Cultural Heuristics

Taking the phenomenon of rush buying Shuanghuanglian as a representative, we expand our discussion by exchanging news reports (possibly true) and rumors related to traditional Chinese culture. Findings show that everyone seems to maintain a supportive or neutral view. Especially among middle-aged people (P4,P6,P7), methods related to traditional Chinese culture proposed by social media are particularly popular and more likely to be trusted:

"During the outbreak of SARS, people have tried the way of fumigating vinegar to kill the virus... So I think fumigating moxibustion, fumigating white vinegar and other related ways can sterilize to a certain extent... In addition to traditional Chinese medicine itself, moxibustion, acupuncture and cupping can also be regarded as the auxiliary physical treatment of traditional Chinese medicine."

What's more, it's worth discussing that there is a saying that "medicine and food have the same origin", which means some food itself is also a kind of traditional Chinese medicine. Ginger, garlic,

vinegar and other ingredients as medicinal materials are very common in traditional Chinese medicine. Therefore, people(P6,P8) also tend to believe in information and reports about daily health food ingredients, even if it is not necessarily correct.

"So...When I read some news about drinking ginger water and eating garlic at home to prevent novel coronavirus, I believe these statements and do what was said."

"Medicinal tea has become an integral part of traditional Chinese medicine. Even if the reported honeysuckle tea and green tea don't really cure or kill the virus, I think they can still help me stay healthy."

Different cultural cognition and health literacy will also affect citizens' attitudes towards judging online public opinion. Li(P12) believes that most people are more inclined to believe the information released by official media:

"It's just human nature... I would be more likely to trust authoritative and official reports, and other sources of information might just be trying to gain attention and profit..."

Wang(P7) suggests that the precise wording of news is also a way to affect his attitude towards information:

"If the headline and content of the news reports use words like maybe, perhaps, or probably, then I assume the information is probably not accurate...Although I also use the word probably in this sentence...hhh"

In terms of special groups, Ding (P8) who is engaged in medical work in hospital has a more professional and rational judgment on information released by social media:

"For news about vaccines, I will check whether they have passed clinical trials and whether there is data support. After all, for most vaccines, research is just beginning, and it's not worth the hype on social media."

5 DISCUSSION

One of the three basic levels of culture embodied in organizational language includes rumor [24]. Rumors are the product of pre-existing culture and information environment which are closely related to group culture [2]. It is also a culture-related public opinion and a lasting form of communication in the global social and cultural landscape, which is closely related to group culture [6]. For example, Lundry addressed rumors are in a specific regional context [17] and Fisher emphasized the importance of "cultural loyalty" [8].

The popularity of social media has changed the way that culture is transmitted. A large number of reports may involve users' cultural backgrounds, thus triggering users' corresponding attitudes towards information. Therefore, we propose the concept of cultural heuristic to consider how cultural factors affect Chinese citizens' response to rumors spread on social media. We explore its role in Chinese citizens' judgment of social media information when facing the public crisis again. As one of the "fast and frugal" heuristics, it provides rules of thumb for the inference of unknown conditions. Findings show that perceptual process and culture background help people understand information and the world. That means something has ever been known or experienced, and thus forms its own cultural or historical context. Then it can easily be identified as having a higher value than the standard.

Through the analysis of the above conversation fragments, it can be found that cultural heuristics plays a significant role in the process of Chinese citizens' judgment of social media reports and information. As a form of cultural communication with a profound cultural origin, rumors inevitably take on cultural characteristics such as history and geography, changing with the development of the times, people's psychology and experience. When facing with uncertain information, individuals often rely on cultural heuristics, which make it easier and more effective to make judgments about complex information. Due to the fact that China is the first country to report COVID-19 news and information on a large scale, it is not easy for the Chinese public to make accurate rumor judgments against the background of unknown risks. The study indicates that individuals may rely on their existing value systems to guide their attitudes and behaviors towards information when faced with uncertainty. In particular, traditional culture represented by traditional Chinese medicine has a significant influence on the attitudes and behaviors of Chinese citizens, especially those who have benefited from traditional Chinese medicine and the elderly.

On the other hand, the research has proved that individuals in Eastern cultures value the public self, tend to believe the oral information of relatives and friends, and follow others' behavior [33]. Collective action frameworks and narratives work when they correspond to the existing cultures and beliefs of the proponents. Therefore, the Eastern group culture encourages people to be more willing to keep pace with the people around them. Peer behavior and group behavior can also influence individuals' attitudes and behaviors towards information. For example, several interviewees have mentioned that they did not plan to buy traditional Chinese medicine, but when they saw that many people around them were buying Chinese medicine online or offline and it was quickly sold out, which can be seen in Figure 2 and Figure 3, they could not help but follow the footsteps of others to buy it, thus changing their initial behaviors. Finally, we also find that the audience's ability to identify rumors is positively correlated with their health literacy from the perspective of cultural cognition. That is, the higher the users' level of health knowledge and culture education level, the more rational they are in the process of rumor identification, and they will have stronger ability to decode rumors in emergencies.

6 CONCLUSION

With COVID-19 has become a serious global problem, false information floods the entire network, the epidemic is changing people's lives and behaviors. Due to the specific history and culture, that is, China has experienced a similar outbreak of SARS virus and the unique cultural background represented by traditional Chinese medicine, we put forward the concept of cultural heuristic. Research selected Chinese citizens as representatives to study whether the attitude towards the COVID-19 rumors is affected by historical experience and cultural background. We use grounded theory to guide our research, and the data comes from observation and semi-structure interview. It enables us to organize interview records and find patterns from the data, so as to obtain a more comprehensive understanding of the research objectives. The results show that cultural heuristic plays a significant role in attitude and behavior when faced with social media information in emergencies. We also find

that individuals usually choose to make a comprehensive judgment of news reports based on fact understanding, cognitive level, the historical and cultural background which they have experienced, and they also have a tendency of collective behavior. We hope that our work will help to understand the source of Chinese public opinion and the cultural motivation behind people's attitudes and behaviors. This work an exploration of rumor culture in non-Western society, and it will help to carry out effective healthy communication and rumor management for the Chinese public in future.

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